Kernix 0.10.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

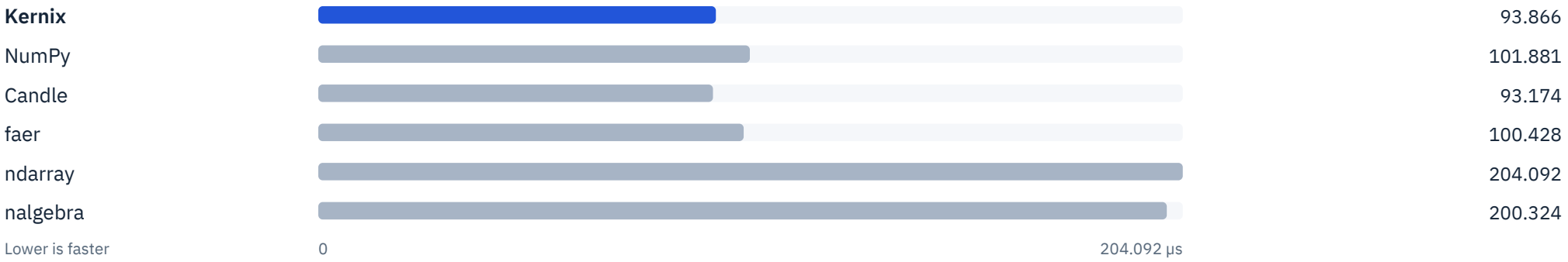
34 workloads

116 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



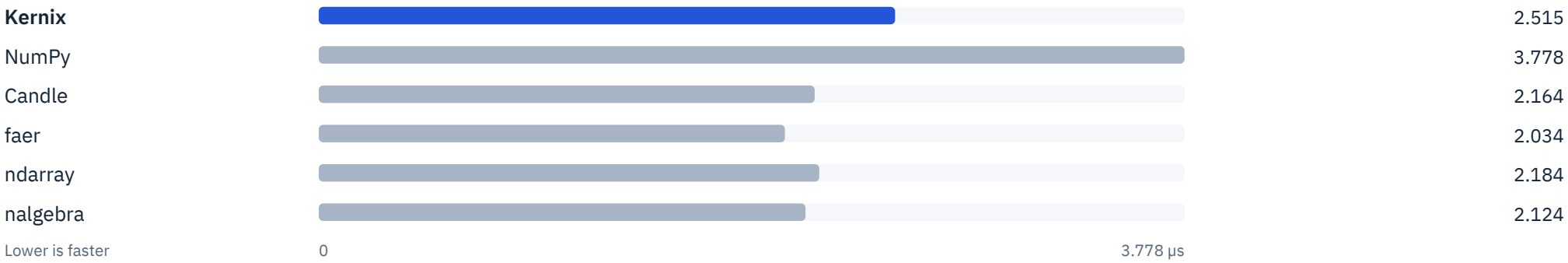
Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

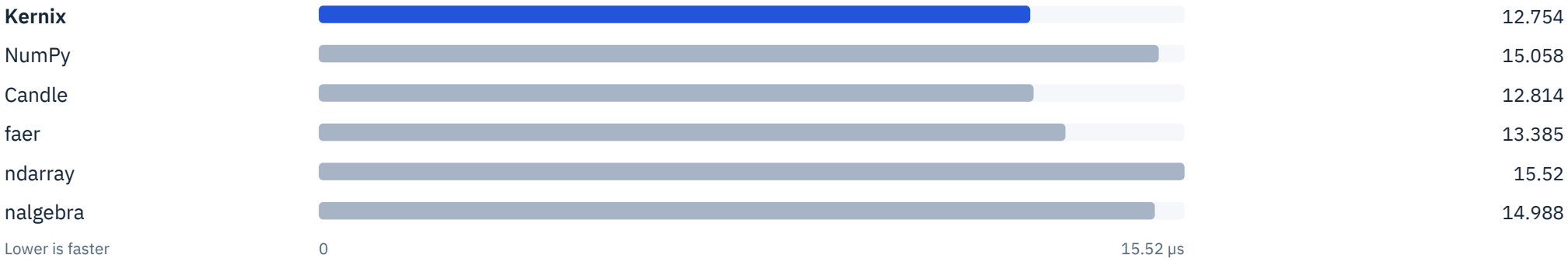
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



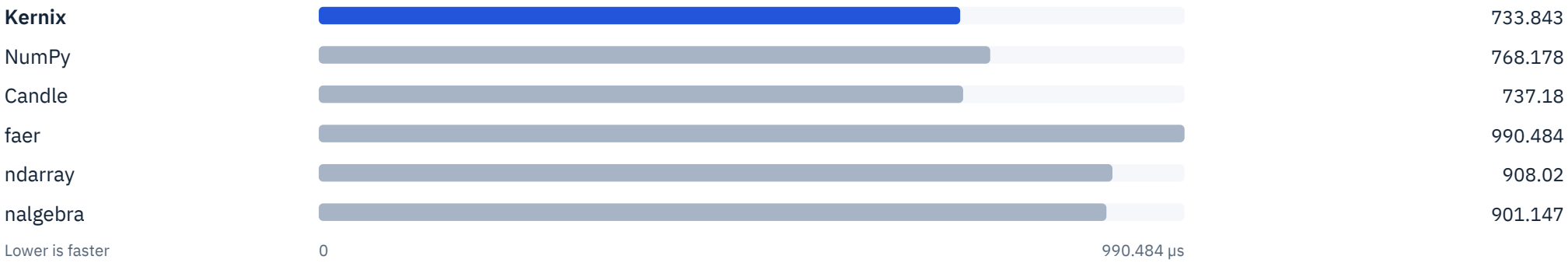
Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

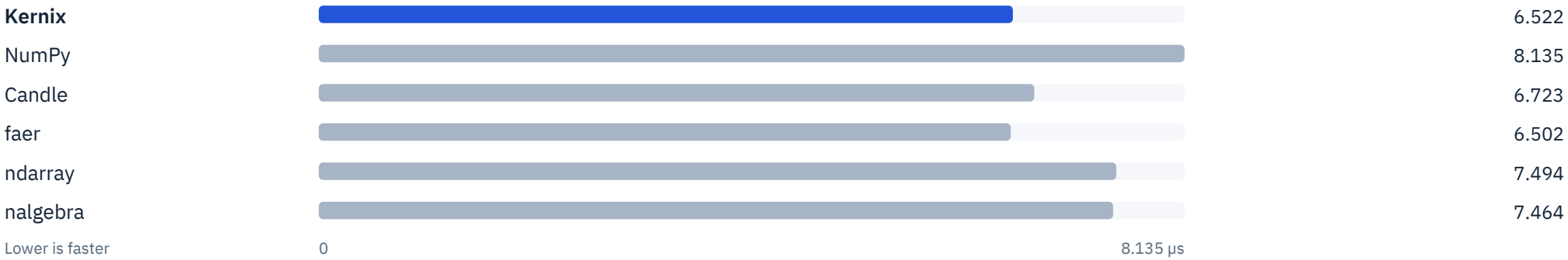
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



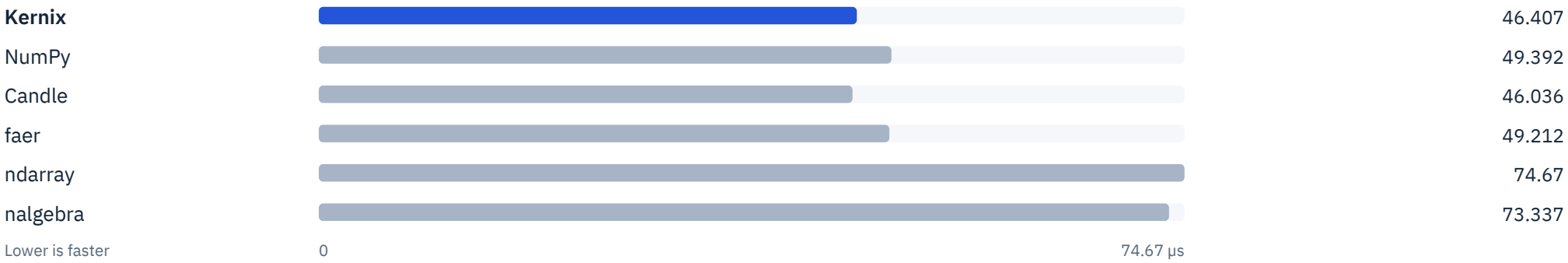
Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

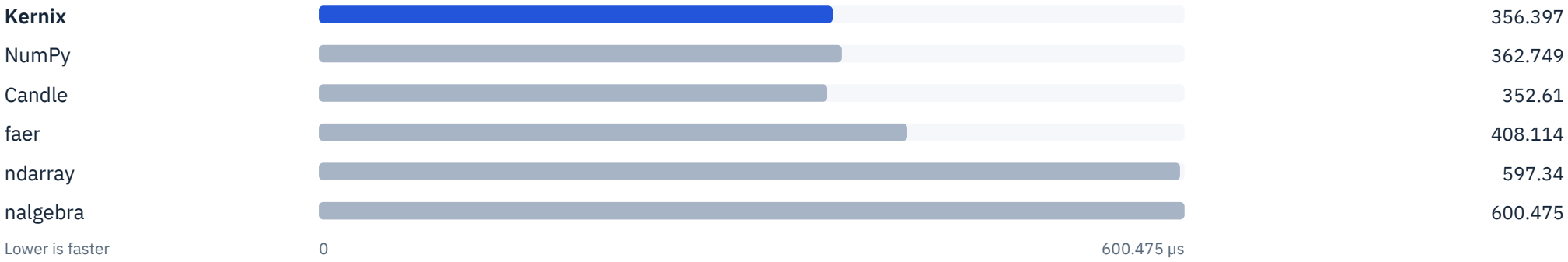
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

Per-call median



Kernix 0.10.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

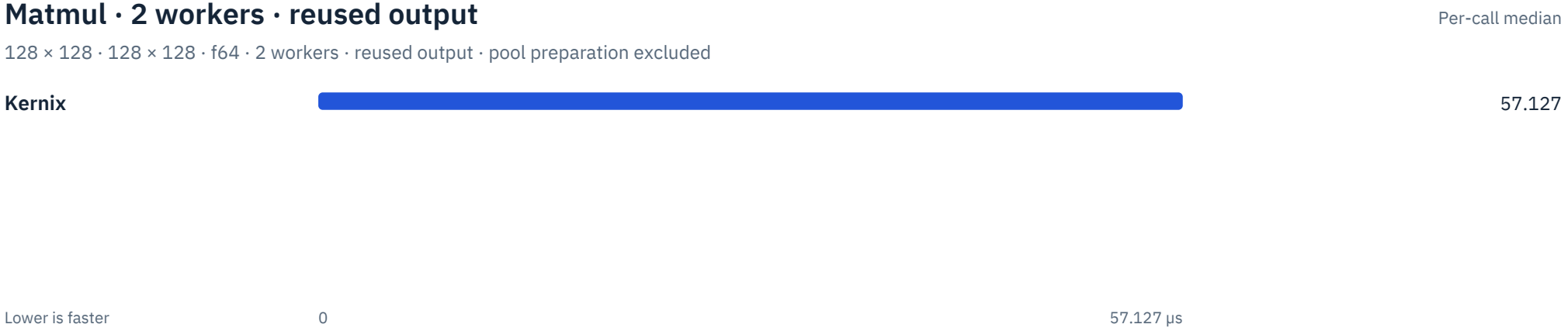
Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.10.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

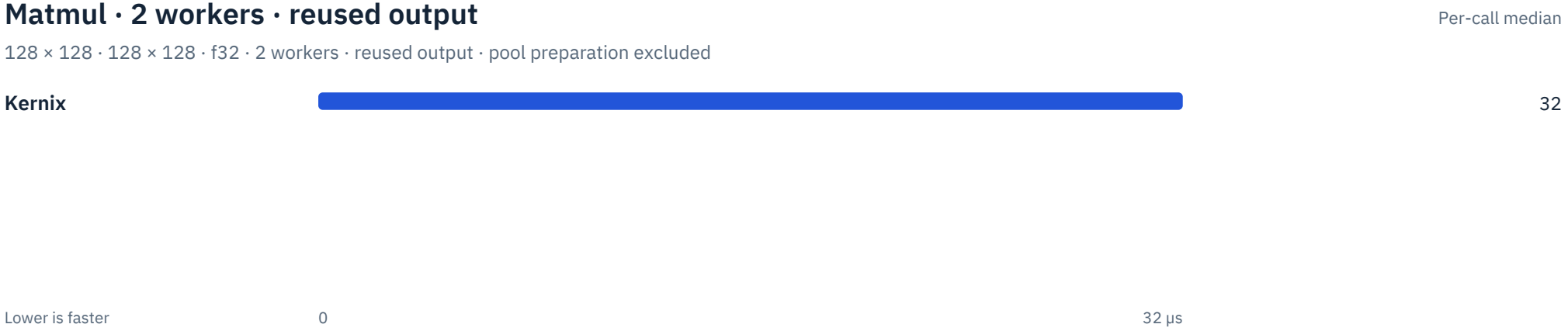
Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



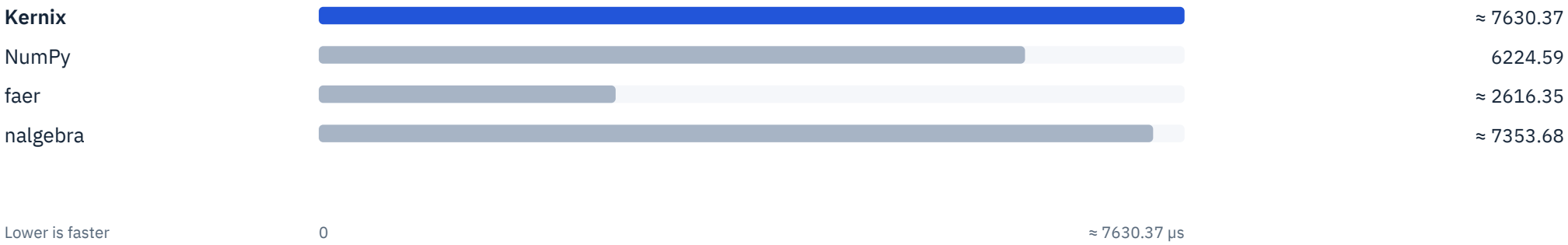
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

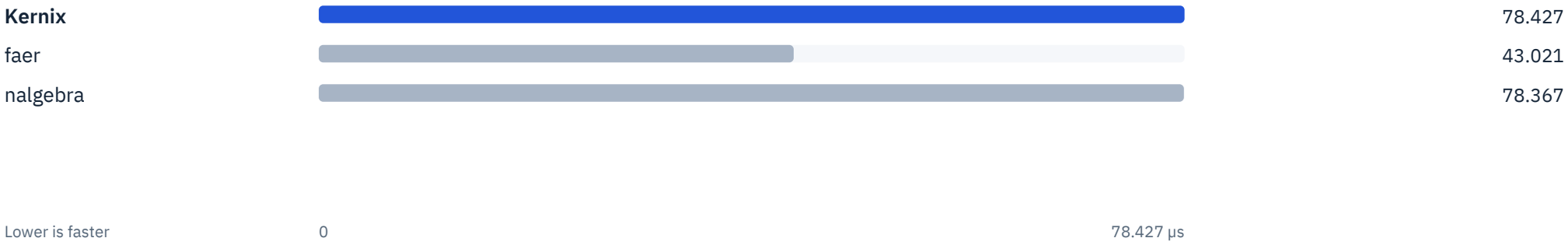
Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



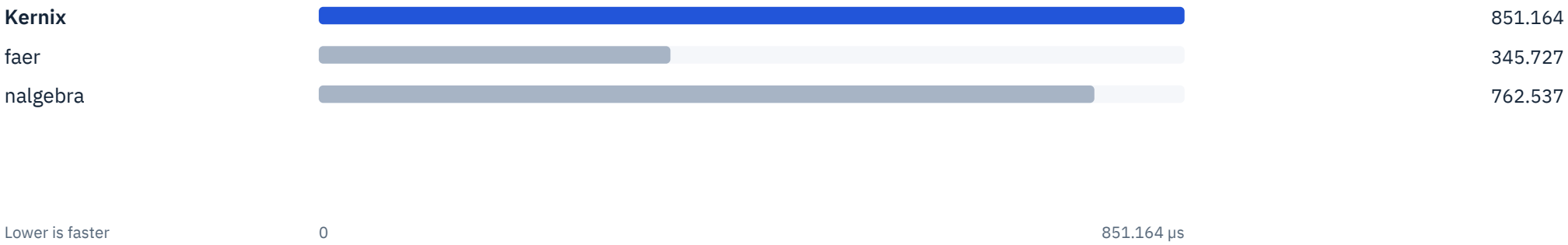
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

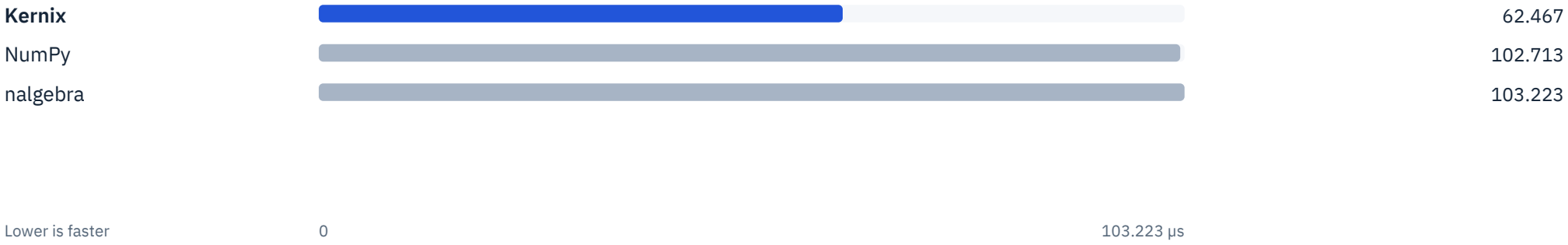
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



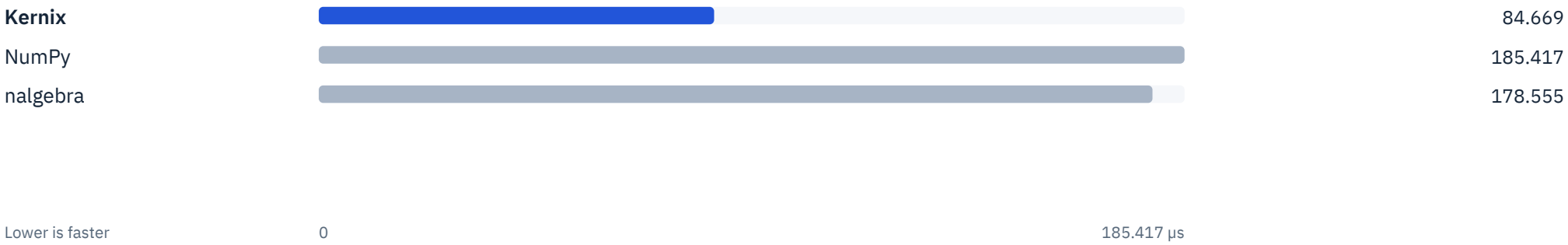
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

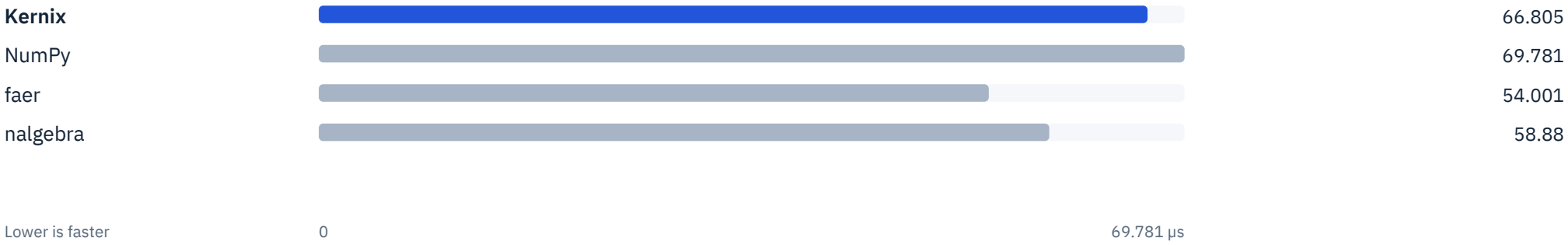
32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



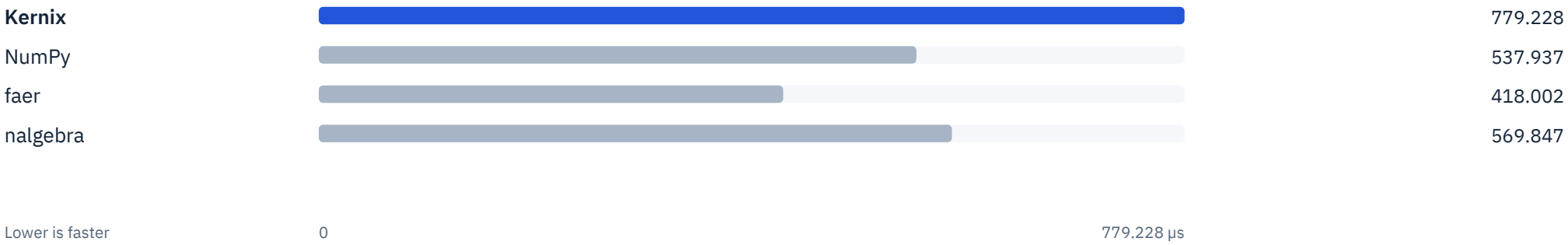
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

256 × 64 · f64 · 1 worker · allocated output

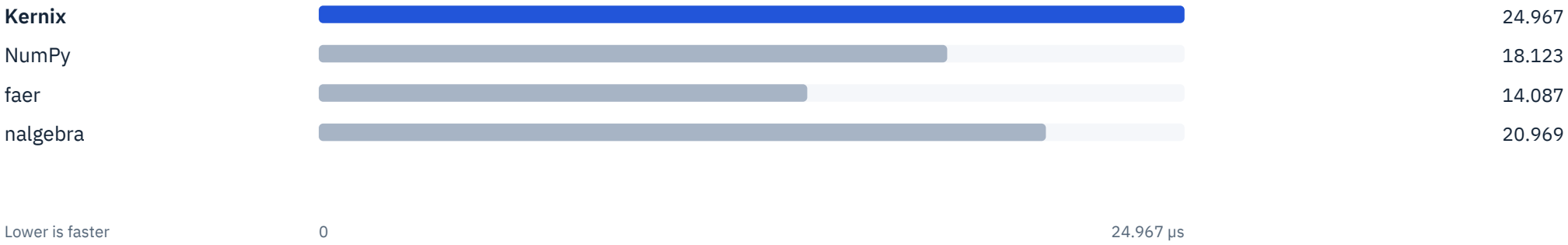
Per-call median



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median



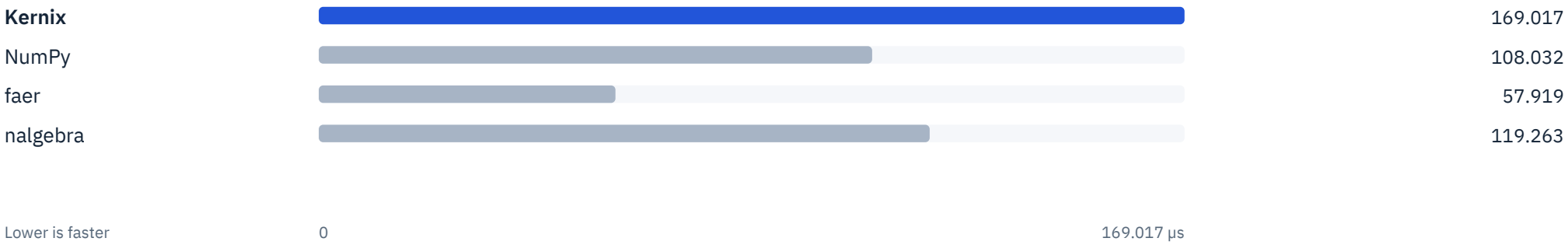
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

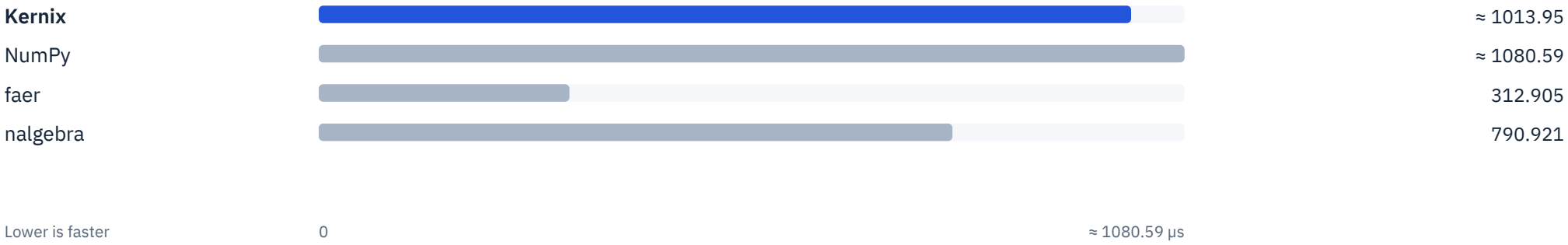
128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



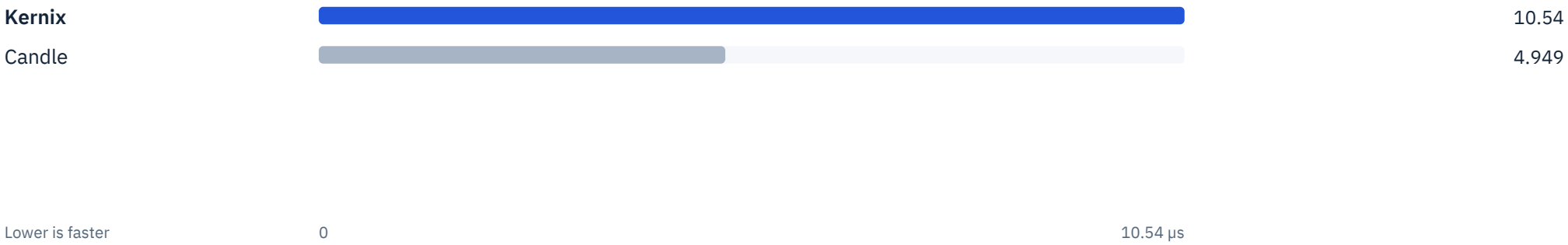
Kernix 0.10.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



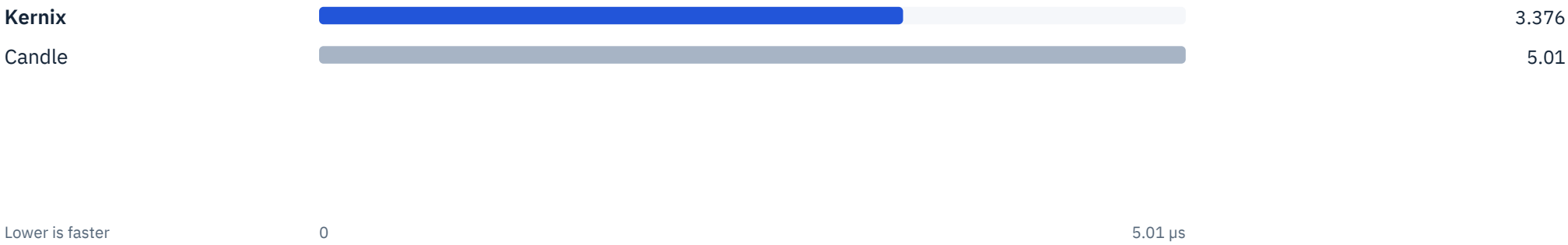
Kernix 0.10.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

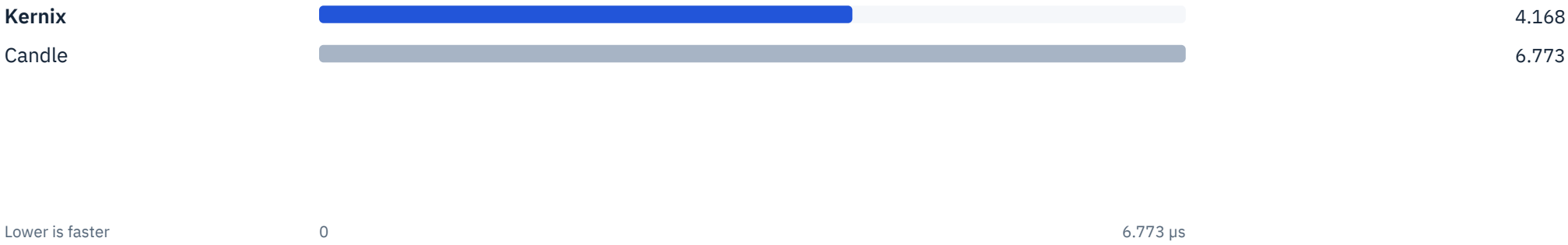
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.10.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

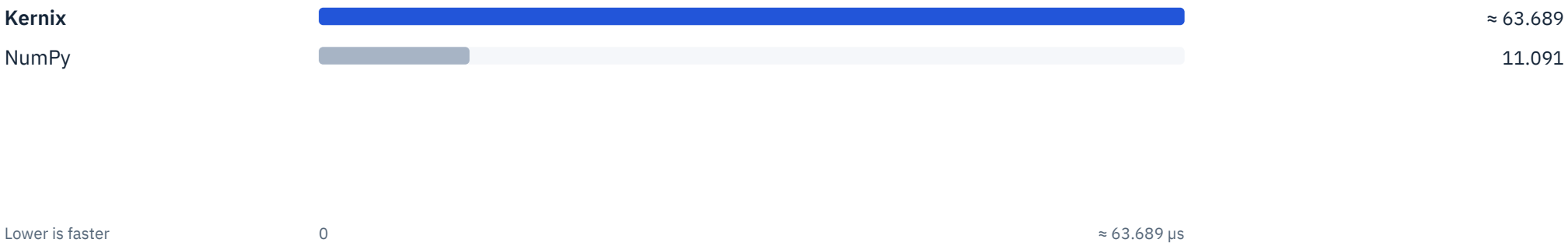


Kernix 0.10.0 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output



Kernix 0.10.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



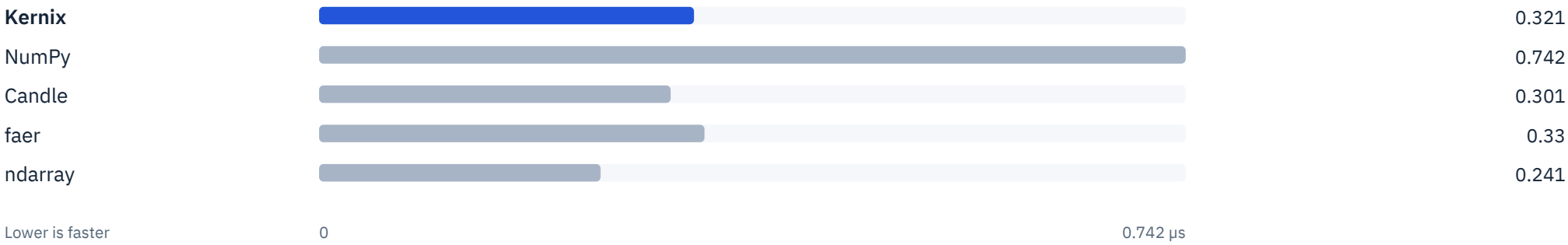
Kernix 0.10.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	93836	93.836
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	102021	102.021
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	49573	49.573
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	46087	46.087
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2164	2.164
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	3778	3.778
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2184	2.184
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2124	2.124
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2034	2.034
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2515	2.515

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	15520	15.52
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	12814	12.814
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	12754	12.754
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	14988	14.988
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	15058	15.058
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	13385	13.385
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	101881	101.881
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	93174	93.174
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	200324	200.324
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	100428	100.428

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	204092	204.092
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	93866	93.866
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	733843	733.843
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	768178	768.178
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	901147	901.147
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	908020	908.02
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	737180	737.18
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	990484	990.484
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7464	7.464
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	6723	6.723

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	6502	6.502
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7494	7.494
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6522	6.522
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	8135	8.135
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	46407	46.407
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	74670	74.67
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	46036	46.036
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	49392	49.392
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	73337	73.337
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	49212	49.212

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	356397	356.397
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	362749	362.749
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	597340	597.34
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	352610	352.61
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	408114	408.114
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	600475	600.475
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3236	3.236
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	4218	4.218
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3076	3.076
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	18845	18.845

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	57127	57.127
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	392625	392.625
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	32000	32
Cholesky factorization / nalgebra 512 × 512 · f64	7353676	7353.676
Cholesky factorization / NumPy 512 × 512 · f64	6224590	6224.59
Cholesky factorization / faer 512 × 512 · f64	2616349	2616.349
Cholesky factorization / Kernix 512 × 512 · f64	7630374	7630.374
Column-pivoted QR / Kernix 96 × 32 · f64	78427	78.427
Column-pivoted QR / faer 96 × 32 · f64	43021	43.021
Column-pivoted QR / nalgebra 96 × 32 · f64	78367	78.367

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 256 × 64 · f64	345727	345.727
Column-pivoted QR / nalgebra 256 × 64 · f64	762537	762.537
Column-pivoted QR / Kernix 256 × 64 · f64	851164	851.164
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	103223	103.223
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	102713	102.713
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	62467	62.467
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	178555	178.555
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	185417	185.417
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	84669	84.669
Thin QR / Kernix 96 × 32 · f64	66805	66.805

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 96 × 32 · f64	54001	54.001
Thin QR / NumPy 96 × 32 · f64	69781	69.781
Thin QR / nalgebra 96 × 32 · f64	58880	58.88
Thin QR / faer 256 × 64 · f64	418002	418.002
Thin QR / Kernix 256 × 64 · f64	779228	779.228
Thin QR / NumPy 256 × 64 · f64	537937	537.937
Thin QR / nalgebra 256 × 64 · f64	569847	569.847
Cholesky factorization / NumPy 64 × 64 · f64	18123	18.123
Cholesky factorization / nalgebra 64 × 64 · f64	20969	20.969
Cholesky factorization / faer 64 × 64 · f64	14087	14.087

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	24967	24.967
Cholesky factorization / Kernix 128 × 128 · f64	169017	169.017
Cholesky factorization / nalgebra 128 × 128 · f64	119263	119.263
Cholesky factorization / faer 128 × 128 · f64	57919	57.919
Cholesky factorization / NumPy 128 × 128 · f64	108032	108.032
Cholesky factorization / faer 256 × 256 · f64	312905	312.905
Cholesky factorization / Kernix 256 × 256 · f64	1013949.0000000001	1013.9490000000001
Cholesky factorization / nalgebra 256 × 256 · f64	790921	790.921
Cholesky factorization / NumPy 256 × 256 · f64	1080593	1080.593
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4949	4.949

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	10540	10.54
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3877.0000000000005	3.8770000000000005
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7715	7.715
Model inference / Candle batched normalized/sliced MLP · f64	5010	5.01
Model inference / Kernix batched normalized/sliced MLP · f64	3376	3.376
Model recording + forward / Kernix batched normalized/sliced MLP · f64	4168	4.168
Model recording + forward / Candle batched normalized/sliced MLP · f64	6773	6.773
Model backward / Kernix prepared graph → fresh gradients · f64	4969	4.969
Model backward / Candle prepared graph → fresh gradients · f64	20859	20.859
Parameter update / Kernix transactional SGD, fixed gradients · f64	802	0.802

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63689.00000000001	63.68900000000001
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	11091	11.091
Row softmax / Candle 256 × 64 · f64	115285	115.285
Row softmax / ndarray 256 × 64 · f64	113523	113.523
Row softmax / Kernix 256 × 64 · f64	119564	119.564
Row softmax / NumPy 256 × 64 · f64	133509	133.509
Create zeros / Kernix 64 × 64 · f32	321	0.321
Create zeros / faer 64 × 64 · f32	330	0.33
Create zeros / ndarray 64 × 64 · f32	241	0.241
Create zeros / Candle 64 × 64 · f32	301	0.301

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	742	0.742
Create ones / Candle 64 × 64 · f32	470	0.47
Create ones / NumPy 64 × 64 · f32	2064	2.064
Create ones / Kernix 64 × 64 · f32	431	0.431
Create ones / ndarray 64 × 64 · f32	351	0.351
Create ones / faer 64 × 64 · f32	391	0.391